

Regression Diagnostics

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Outline

- 1 Residual Analysis and Diagnostics
- 2 Regression Performance Measures

Residual Analysis and Diagnostics

Learning outcomes

- State the assumptions behind least squares inference and explain how residuals reveal violations
- Compute and interpret standardized residuals and leverage from the hat matrix
- Diagnose and remedy non-linearity, correlated errors, non-constant variance, outliers, high-leverage observations and collinearity
- Compute and interpret MSE, RMSE, MAE, RSE, R^2 and adjusted R^2
- Distinguish training error from test error

Residuals and model assumptions

- Residual vector $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ with entries $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$; least squares gives $\tilde{\mathbf{X}}^\top \mathbf{e} = \mathbf{0}$, so the residuals sum to zero and are uncorrelated with every feature
- Standard errors, confidence and prediction intervals, and tests rely on assumptions about the errors: the relationship is linear in the coefficients, the errors $\varepsilon^{(i)}$ are uncorrelated, they have constant variance σ^2 , and (for exact small-sample inference) they are normal
- **Residual analysis** checks these assumptions with plots; the main tool is the plot of residuals against fitted values: random scatter around zero is good, a curve signals non-linearity, a funnel signals non-constant variance
- Three further problems: outliers, high-leverage observations, and collinearity

Residuals and the hat matrix

- Fitted values and residuals are linear functions of the labels: $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$, with the **hat matrix** $\mathbf{H} = \tilde{\mathbf{X}}(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top$ (symmetric, idempotent, $\text{tr}(\mathbf{H}) = p + 1$)
- Since $(\mathbf{I} - \mathbf{H})\tilde{\mathbf{X}} = \mathbf{0}$, the residuals depend on the errors only: $\mathbf{e} = (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}$
- Hence $\mathbb{E}[\mathbf{e}] = \mathbf{0}$ and $\text{Cov}(\mathbf{e}) = \sigma^2(\mathbf{I} - \mathbf{H})$; reading off the diagonal,
$$\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)}), \quad h^{(i)} = H_{ii}$$
- Even when all errors have the same variance, the residuals do not: observations with large **leverage** $h^{(i)}$ have residuals with smaller variance, because the fit is pulled towards them; the leverages add up to $\sum_i h^{(i)} = p + 1$

Standardized (studentized) residuals

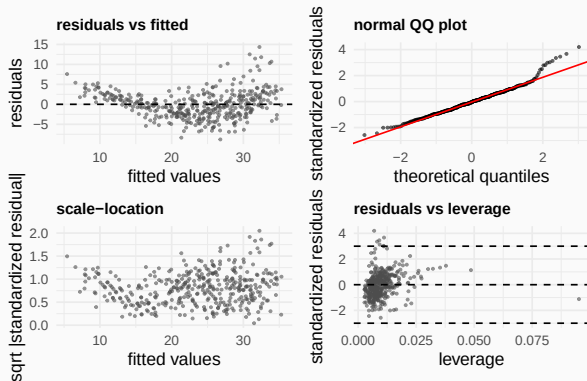
- Raw residuals have unequal variances and carry the units of y ; dividing by the estimated standard deviation fixes both:

$$r^{(i)} = \frac{e^{(i)}}{\hat{\sigma}\sqrt{1 - h^{(i)}}}, \quad \hat{\sigma} = \text{RSE}$$

- Each $r^{(i)}$ has variance close to 1 and no units, so residuals can be compared across observations and across models
- With normal errors about 99.7% of the $r^{(i)}$ fall within ± 3 : an observation with $|r^{(i)}| > 3$ is unusual, but among 400 observations one such value is expected by chance
- ISLP calls $r^{(i)}$ the studentized residual; in R it is computed by `rstandard()`

Four standard diagnostic plots

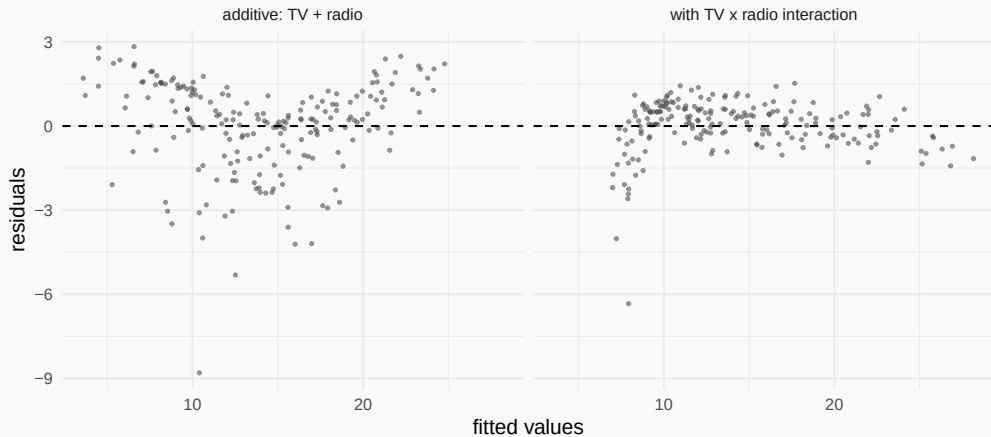
- **Residuals vs fitted:**
curve = non-linearity,
funnel = non-constant
variance
- **Normal QQ plot:** points
near the line = roughly
normal errors
- **Scale-location:** upward
trend = non-constant
variance
- **Residuals vs leverage:**
flags unusual
observations
- **Shapiro-Wilk test:** p-value



Problem 1: non-linearity

- If the true relationship is non-linear, predictions and inference are unreliable; the residual plot shows a pattern, often a U-shape
- Advertising, additive TV + radio model: positive residuals at the smallest and largest fitted values, negative in between, a sign of missing synergy
- With the TV $\times$ radio interaction the curvature weakens and RSE falls from 1.68 to 0.94; a few large negative residuals remain
- Remedies: transform features ($\log x$, $\sqrt{x}$), add interaction terms, or fit a more flexible model

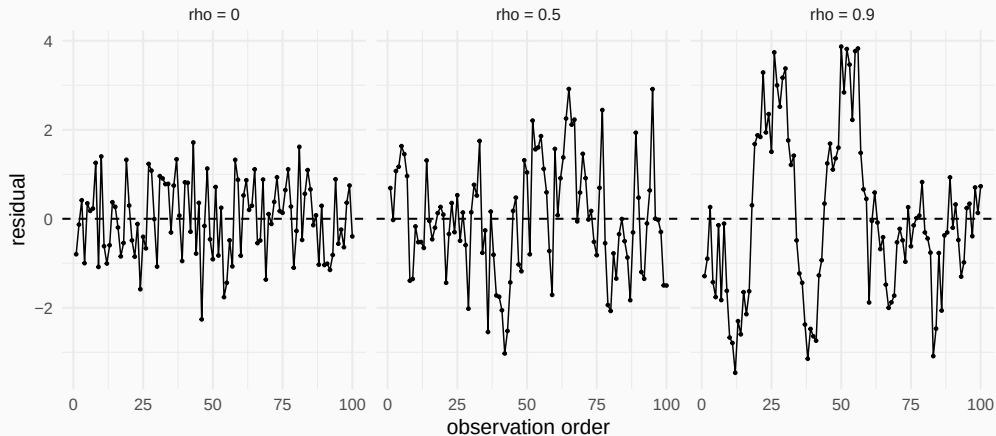
Problem 1: non-linearity (cont.)



Problem 2: correlated errors

- The errors are assumed uncorrelated; this fails for time series (neighbouring errors alike) and for repeated measurements on the same unit
- Consequence: standard errors are too small, so intervals are too narrow and p -values too small
- Detection: plot residuals in observation order; tracking (neighbouring residuals alike) signals positive correlation (ρ = correlation of successive errors)
- Remedy: time-series or mixed models (out of the scope this lecture)

Problem 2: correlated errors (cont.)



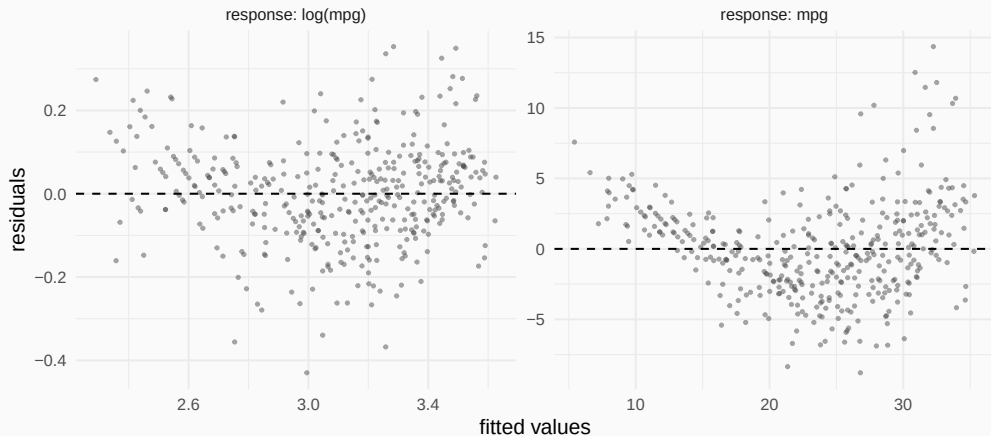
Problem 3: non-constant variance

- **Heteroscedasticity:** $\text{Var}(\varepsilon^{(i)})$ changes with the level of the response (a funnel in the residual plot), making standard errors, intervals and tests unreliable
- Auto, mpg on horsepower, weight and year: the residual standard deviation grows from 2.35 to 2.48, 3.47 and 3.88 across the four quartiles of the fitted values
- Remedy 1, transform the response ($\log y$ or $\sqrt{y}$; concave transformations shrink large values more): for $\log(\text{mpg})$ the spread is nearly constant (0.116, 0.117, 0.130, 0.107); coefficients then describe multiplicative effects

Problem 3: non-constant variance (cont.)

- Remedy 2, weighted least squares: minimize $\sum_i w^{(i)} (e^{(i)})^2$ with weights $w^{(i)} \propto 1/\text{Var}(\varepsilon^{(i)})$, giving $\hat{\beta} = (\tilde{\mathbf{X}}^\top \mathbf{W} \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{W} \mathbf{y}$ with $\mathbf{W} = \text{diag}(w^{(i)})$

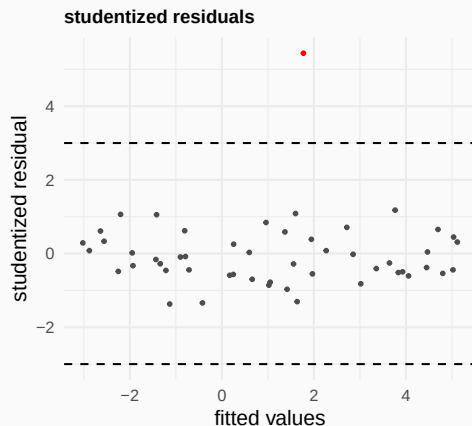
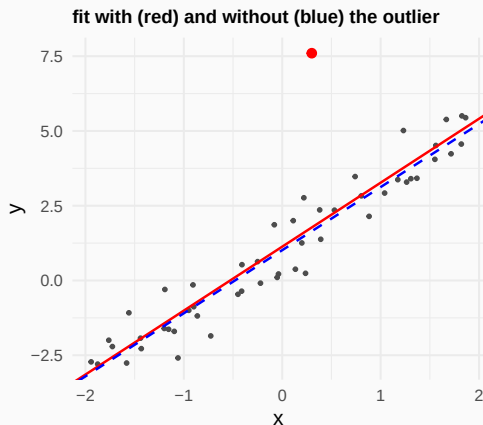
Problem 3: non-constant variance (cont.)



Problem 4: outliers

- An **outlier** is an observation whose $y^{(i)}$ is far from the fitted value (a recording error or a genuinely unusual case)
- At a typical x it hardly moves the line, but it inflates RSE and lowers R^2
- Flag observations with $|r^{(i)}| > 3$, using the standardized residual $r^{(i)} = e^{(i)} / (\hat{\sigma} \sqrt{1 - h^{(i)}})$
- Investigate before deleting; Auto fit: 6 of 392 cars have $|r^{(i)}| > 3$, and the largest ($r^{(i)} = 4.20$) is the most economical car in the data (46.6 mpg), far above its prediction

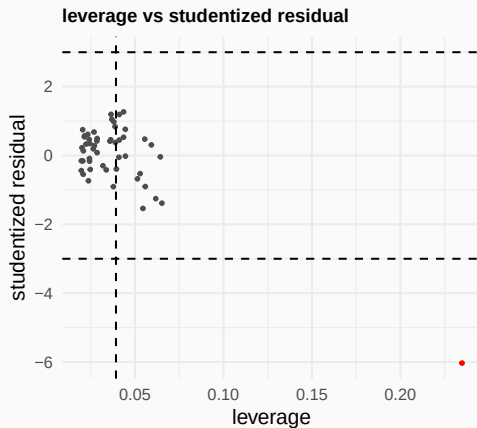
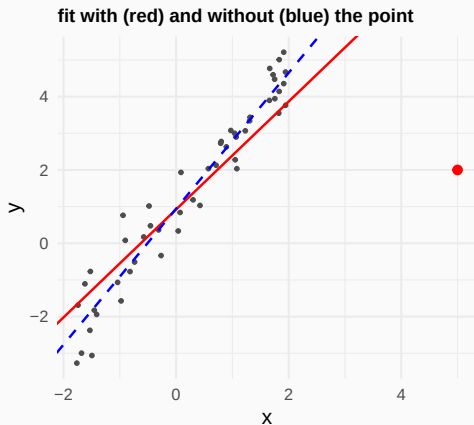
Problem 4: outliers (cont.)



Problem 5: high-leverage observations

- A **high-leverage** observation has unusual features $\mathbf{x}^{(i)}$ and pulls the fit towards itself; leverage $h^{(i)}$ depends only on the features, never on $y^{(i)}$
- $1/n \leq h^{(i)} \leq 1$ with average $(p+1)/n$; values far above the average signal high leverage, and with a large residual the point is *influential*: removing it changes the fit a lot
- Auto fit: the largest leverage is $h^{(i)} = 0.095$ (a 225-horsepower car), 9.3 times the average $4/392 = 0.0102$

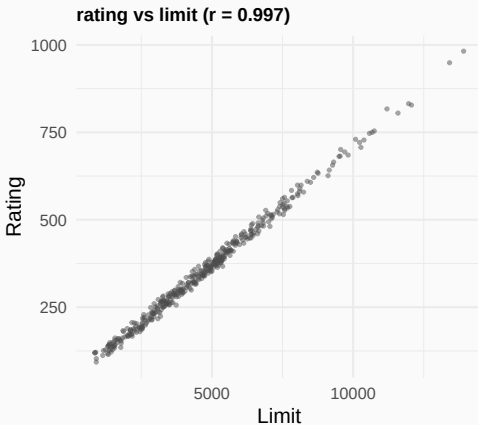
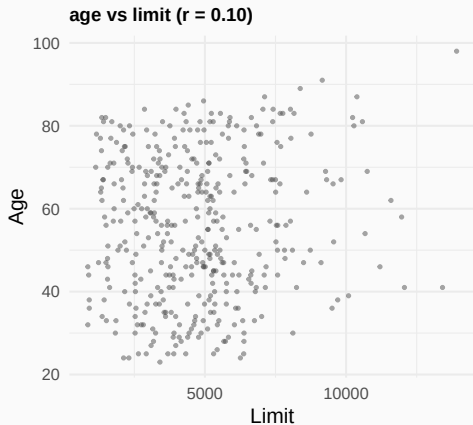
Problem 5: high-leverage observations (cont.)



Problem 6: collinearity

- **Collinearity:** features are highly correlated, so their separate effects on y cannot be told apart ($\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}}$ is nearly singular)
- Credit: limit and rating have $r = 0.997$, limit and age only $r = 0.10$
- Consequences: unstable coefficients and inflated standard errors; an important feature can look insignificant
- Balance on age + limit: $\text{SE}(\hat{\beta}_{\text{limit}}) = 0.0050$, $p < 0.0001$; on rating + limit: 0.0638 (13 times larger), $p = 0.70$

Problem 6: collinearity (cont.)



Detecting collinearity: variance inflation factor

- Pairwise correlations can miss collinearity among three or more features (multicollinearity); the **variance inflation factor** does not

$$\text{VIF}(\hat{\beta}_j) = \frac{1}{1 - R_{x_j|x_{-j}}^2}$$

- $R_{x_j|x_{-j}}^2$ is the R^2 from regressing feature x_j on all the other features; VIF is at least 1, and values above 5 or 10 signal a problem
- Credit, age + rating + limit: VIF = 1.01 (age), 160.7 (rating), 160.6 (limit); with only age and limit: 1.01 and 1.01
- Remedies: drop one of the redundant features, or combine them (for example, average the standardized features)
- Predictions barely change ($R^2 = 0.750$ and 0.746 for the two models), but individual coefficients and their inference do

Diagnostic checklist

problem	symptom	remedy
non-linearity	curve in residuals vs fitted	transform features, interact
correlated errors	tracking of residuals over time	time-series or mixed model
non-constant variance	funnel in residuals vs fitted	transform y , weighted least
outliers	$ r^{(i)} > 3$	investigate, fix or remove e
high leverage	$h^{(i)}$ far above $(p+1)/n$	check the point, refit without
collinearity	VIF above 5 or 10	drop or combine features

- Check in this order: residual plots first, then unusual observations, then collinearity; every fix changes the model, so refit and repeat the diagnostics

Regression Performance Measures

Measures of the quality of fit

measure	definition	comment
MSE	$\frac{1}{n} \sum_{i=1}^n (y^{(i)} - \hat{y}^{(i)})^2 = \text{RSS}/n$	squared units of y
RMSE	$\sqrt{\text{MSE}}$	units of y
MAE	$\frac{1}{n} \sum_{i=1}^n y^{(i)} - \hat{y}^{(i)} $	units of y , less sensitive to outliers
RSE	$\sqrt{\text{RSS}/(n - p - 1)}$	units of y , estimates σ
R^2	$1 - \text{RSS}/\text{TSS}$	unit-free, never falls with more features
adjusted R^2	$1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)}$	unit-free, penalizes each added feature

- MSE and RMSE punish large errors heavily (squares); MAE weighs all errors linearly, so a few outliers move it less

RSE versus RMSE: why divide by $n - p - 1$?

- RMSE averages the squared residuals over n ; RSE divides by the degrees of freedom $n - p - 1$ instead, because $p + 1$ coefficients were fitted to the same data
- $\mathbb{E}[\text{RSS}] = (n - p - 1)\sigma^2$, so $\text{RSE}^2 = \text{RSS}/(n - p - 1)$ is unbiased for σ^2 , while $\text{MSE} = \text{RSS}/n$ underestimates it by the factor $(n - p - 1)/n$
- Advertising, TV + radio + newspaper: $\text{RSS} = 556.83$ gives $\text{MSE} = 2.78$ ($\text{RMSE} = 1.669$) but $\text{RSE}^2 = 2.84$ ($\text{RSE} = 1.686$)
- The gap grows with the number of features: with many features relative to n , the residuals look much smaller than the true errors, an early sign that in-sample fit is optimistic

Interpreting R^2

- $R^2 = 1 - \text{RSS}/\text{TSS} = \text{ESS}/\text{TSS}$ is also the squared correlation between y and $\hat{y}$, and equals r^2 in simple regression
- It lies in $[0, 1]$ for a model with an intercept, and it never decreases when a feature is added: the larger model can always reproduce the smaller one by setting the new coefficient to zero
- It depends on the spread of the data: the same model fitted where x (and hence y) varies over a wider range has a higher R^2 , and R^2 values of models with different responses (for example mpg and log mpg) are not comparable
- High R^2 does not mean the model is right (check the residuals); low R^2 does not mean it is useless, since real effects can be small relative to noise, for example in survey or behavioural data

Adjusted R-squared

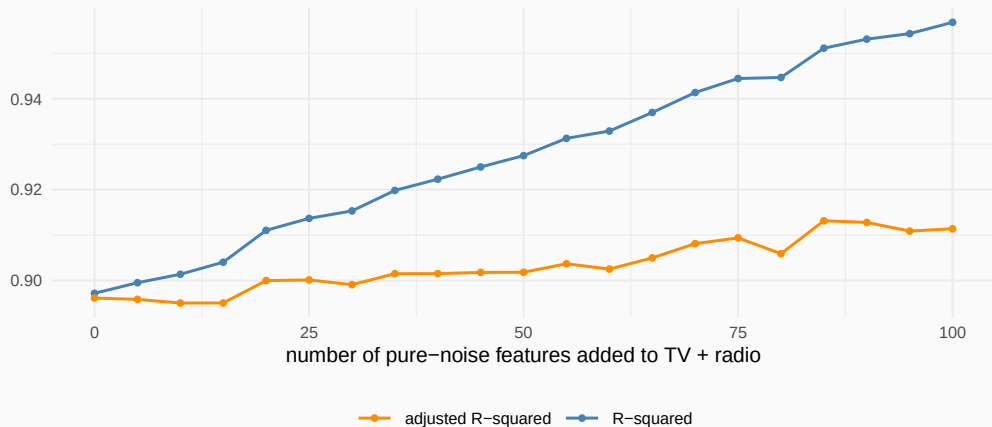
$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)} = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

- It compares two unbiased variance estimates: the residual variance $\hat{\sigma}^2$ and the variance of y_i ; the factor $(n - 1)/(n - p - 1)$ grows with p and penalizes each added feature
- Adding a feature raises R_{adj}^2 exactly when the feature's $|t|$ -statistic exceeds 1 (equivalently, its partial F exceeds 1); a weak bar, but it stops R_{adj}^2 from rising for pure noise
- Advertising: adding newspaper has $|t| = 0.18 < 1$, so R_{adj}^2 falls from 0.8962 to 0.8956 while R^2 stays 0.8972

Adjusted R-squared with useless features

- Figure: pure-noise features are added one block at a time to the TV + radio model
- R^2 can only rise, here from 0.897 to about 0.95 with 100 noise features; R^2_{adj} stays near 0.90 and shows no systematic gain

Adjusted R-squared with useless features (cont.)



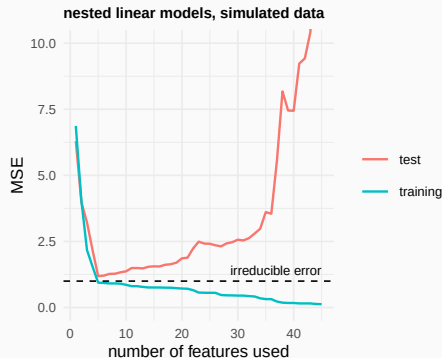
Comparing regression models on the Advertising data

model	RSE	RMSE	MAE	R^2	adjusted R^2
TV	3.259	3.242	2.550	0.6119	0.6099
TV + radio	1.681	1.669	1.254	0.8972	0.8962
TV + radio + newspaper	1.686	1.669	1.252	0.8972	0.8956
TV $\times$ radio	0.944	0.934	0.660	0.9678	0.9673

- RMSE and MAE are in thousands of units of sales: the typical error falls from about 3.2 (TV only) to about 0.9 (TV $\times$ radio)
- Adding newspaper leaves R^2 unchanged (0.8972) but lowers adjusted R^2 (0.8962 to 0.8956) and raises RSE: the penalty exposes a useless feature
- All numbers here are computed on the training data used to fit the models

Training error versus test error

- **Training MSE:** $\frac{1}{n} \sum_i (y^{(i)} - \hat{f}(\mathbf{x}^{(i)}))^2$
on the data used to fit $\hat{f}$; all earlier measures are training measures
- **Test MSE:** the same average over new observations not used in fitting
- More features always lower the training MSE; once the informative ones are in, the rest fit noise and the test MSE rises: **overfitting**
- Simulation: 50 observations, 45 features, 5 informative



Summary

- **Residuals:** $\mathbf{e} = (\mathbf{I} - \mathbf{H})\boldsymbol{\epsilon}$ with $\text{Var}(e^{(i)}) = \sigma^2(1 - h^{(i)})$; standardized residuals $r^{(i)} = e^{(i)} / (\hat{\sigma}\sqrt{1 - h^{(i)}})$
- **Diagnostics:** residual plots reveal non-linearity, correlated errors and non-constant variance; standardized residuals flag outliers, leverage flags unusual feature values, VIF flags collinearity
- **Performance:** MSE, RMSE, MAE, RSE, R^2 and adjusted R^2 measure the fit to the training data; RSE and adjusted R^2 account for the degrees of freedom
- **Training vs test:** training error keeps falling as features are added, test error does not

Questions?
